

Convexity (MCQ/TF every year)

Def. f convex $\iff$ epi(f) convex $\iff f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \ \forall \theta \in [0, 1]$.

Characterizations (use these):

- Twice diff.: $\nabla^2 f(x) \succeq 0 \ \forall x$ (eigenvalues for quadratics $x^\top Q x$).
- 1st-order: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) \ \forall x, y$.
- Monotone grad: $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0$.
- $f^{**} = f$ (proper, closed epi).
- Restriction to line convex.

Preserving: nonnegative lin. comb., pointwise max, affine comp., intersection of sets.

Does *NOT* imply convex (traps):

- Unique/global minimizer or single local min (counterex: $f(x) = |x|$ on $[-1, 1]$, 1 outside).
- $\nabla^2 f$ bounded below.
- $|\nabla f(x) - \nabla f(y)| \geq \mu |x - y| \ (\mu > 0)$.
- Pointwise min of convex funcs (counterex: $(x + 1)^2, (x - 1)^2$).
- Product $f_1 f_2$ (counterex: $x \cdot (-x) = -x^2$).
- $g(x) = \exp(f(x))$ if f convex (not always).
- $\sqrt{x}$ on $[0, 1]$: no finite Lip. const. ($\nabla f \rightarrow \infty$ at 0).

Examples:

- $\frac{1}{2}x^2 + p \ln x$ on $[1, \infty)$: convex $\iff p \leq 1$ ($\nabla^2 = 1 - p/x^2 \geq 0$).
- $1/x_i$ convex on $\{x > 0\}$.
- $f(x) = 1/\ln x \ (x > 1)$: check Hessian (not convex).

Tactic (MC/TF): Test Hessian/1st-order or give counterex. Local min = global $\implies$ convex? **False**.

Conjugate $f^*(y) = \sup_x (y^\top x - f(x))$

f proper convex closed $\implies f^{**} = f. \ y \in \partial f(x) \iff x \in \partial f^*(y)$.

Compute: Maximize $y x - f(x)$; stationary pt. (deriv=0) if in dom; else boundaries. Cases needed.

Quiz Examples:

- $f(x) = -x^2, x \in [-1, 1]$: $f^*(y) = 1 + |y|$ (V-shape).
- $f(x) = -\ln x \ (x > 0)$: $f^*(y) = -1 - \ln(-y) \ (y < 0), \infty$ else.
- $f(x) = e^x$: $f^*(y) = y \ln y - y \ (y > 0), \infty$ else.
- $f(x) = x^2 + x, x \in [-1, 1]$:

$$f^*(y) = \begin{cases} -y & y \leq -1 \\ \frac{1}{4}(y - 1)^2 & -1 \leq y \leq 3 \\ y - 2 & y \geq 3 \end{cases}$$

(Stationary $x = \frac{1}{2}(y - 1)$; check interval).

Tactic: Plot asked (label axes). Stationary often quadratic result; evaluate boundaries.

Subdifferentials & Optimality

$\partial f(\bar{x}) = \{v \mid f(x) \geq f(\bar{x}) + v^\top (x - \bar{x}) \ \forall x\}$.

Optimality: $0 \in \partial f(x^*) \iff x^*$ global min.

Example: $f(x) = |x|$:

$$\partial f(x) = \begin{cases} \{-1\} & x < 0 \\ [-1, 1] & x = 0 \\ \{1\} & x > 0 \end{cases}$$

Minimizer where $0 \in \partial f$ (interval includes 0).

Tactic (plot questions): Find point where subdiff plot contains 0.

Duality, Lagrangian, KKT

Primal: $\min f(x)$ s.t. $g_i(x) \leq 0, h_j(x) = 0$.

Lagrangian: $\mathcal{L}(x, \lambda, \nu) = f(x) + \sum \lambda_i g_i(x) + \sum \nu_j h_j(x), \lambda \geq 0$.

Dual fn: $d(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$.

Weak duality: $d^* \leq p^*$ **always**.

Strong duality + KKT suff./nec. (convex + Slater: $\exists \bar{x}, g_i(\bar{x}) < 0$):

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) &= 0 \\ g(x^*) &\leq 0, \ h(x^*) = 0, \ \lambda^* \geq 0 \\ \lambda^{*\top} g(x^*) &= 0 \text{ (compl. slackness)} \end{aligned}$$

2023 SVM ex.: $\min \frac{1}{2} \|a\|^2$ s.t. $y_i(a^\top x_i + b) \geq 1$. Dual $\max_{\lambda \geq 0} d(\lambda)$. For $x_A = (0, 0)^\top, y_A = -1, x_B = (2, 2)^\top, y_B = 1$: $\lambda_A^* = \lambda_B^* = 1/4, a^* = (1/2, 1/2)^\top, b^* = -1$.

2024 ex.: $f = 1/x_1 + 4/x_2, x_1 + x_2 \leq 3, x_{1,2} \geq 1$: write $\mathcal{L}$, solve KKT stationarity+slack $\rightarrow x^*, p^*$.

Tactic: Write full $\mathcal{L}$, set $\nabla_x \mathcal{L} = 0$, substitute into $d(\lambda)$, maximize (often quadratic). Check Slater.

Problem Classes (Hierarchy)

$LP \subset QP \ (P \succeq 0) \subset QCQP \subset SOCP \subset SDP$.

2024 Trap: $\min f^\top x$ s.t. $|A_i x + b_i|^2 \leq c_i^\top x + d_i, Fx = g$:

- SOCP: yes ($\|A_i x + b_i\|_2 \leq c_i^\top x + d_i$).
- Not always QP/SDP (quad constr. not quadratic form).

Smoothness, Strong Conv., Rates

μ -strongly convex + L -smooth $\implies \kappa = L/\mu$.

Defs:

- L -smooth: $\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \iff \nabla^2 \preceq L I$.
- μ -str. conv.: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|x - y\|^2 \iff \nabla^2 \succeq \mu I$.
- Lip. convex (f Lip. const. L_f): subgrad. method.

Convergence ($f(x_N) - f(x^*) \leq \varepsilon$):

- μ -sc, L -sm GD: $N \sim \kappa \ln(1/\varepsilon)$.
- Nesterov acc. ($\mu > 0$): $N \sim \sqrt{\kappa} \ln(1/\varepsilon)$.
- L -sm convex (no μ): Nesterov $N \sim \sqrt{LR^2/\varepsilon} \ (O(1/\sqrt{\varepsilon}))$.
- Lip. convex (subgrad.): $N \sim (RL_f/\varepsilon)^2 \ (O(1/\varepsilon^2))$.
- Projected GD: same rate as GD (if proj. cheap: ℓ_1 -ball, simplex, ℓ_∞).

Tactic (rates): Match assumptions exactly. $\sqrt{x}$ on $[0, 1]$: Lip. const. does **not exist**.

Tactics & Formulas

MC/TF Convexity: Hessian or counterex. (use above). $f = f^{**}$ needs proper+closed epi.

Conjugate/KKT: Explicit deriv + boundary cases; compl. slackness key.

Plots: Hyperplane through midpoint (SVM); V-shape for $1 + |y|$; label axes.

Matrix: $\nabla(x^\top A x) = (A + A^\top)x, \nabla^2 = A + A^\top$.

Quadratic $f(x) = \frac{1}{2} x^\top Q x + b^\top x$: min at $-Q^{-1}b$ if $Q \succ 0$.

General Tips:

- Weak duality always. Strong + KKT under Slater.
- $\partial f(0) = [-1, 1]$ for $|x|$.
- Check domains (e.g. $x > 0$ for $\ln$).
- One A4: prioritize convexity tests, conjugate examples, KKT steps, rate table.

Common funcs (memorize conjugates): $\pm x^2$ (restricted), $\ln$, $\exp, 1/x_i, x^2 + x$, norms.

1. Convexity (Quiz 1 Tactic)

Basic Tests:

- Def:** $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \ \forall \theta \in [0, 1] \iff$ epi(f) convex.
- 1st-order:** $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$. Monotone grad: $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0$.
- 2nd-order:** $\nabla^2 f(x) \succeq 0$. (Strict if $\succ 0$).
- Preserving:** Non-negative sum, affine comp. $f(Ax + b)$, pointwise max $g(x) = \max_i f_i(x)$, intersection of sets. $f^{**} = f$ (proper, closed).

The "NOT CONVEX" Trap List:

- Pointwise **min** of convex funcs (e.g., $(x + 1)^2, (x - 1)^2$).
- Unique/global min $\nRightarrow$ convex (e.g., $f(x) = |x|$ on $[-1, 1]$, 1 outside).
- $\nabla^2 f$ bounded below $\nRightarrow$ convex (must be ≥ 0).
- Product $f_1 \cdot f_2$ of convex $\nRightarrow$ convex (e.g., $x \cdot (-x) = -x^2$).
- $1/f(x)$ or $\exp(f(x))$ where f is convex (not always).
- $f(x) = 1/\ln x$ on $(1, \infty)$ (Check Hessian: $x = e^2 \rightarrow \text{neg}$).

2. Conjugates $f^*(y)$

Definition: $f^*(y) = \sup_x \{y^\top x - f(x)\}$. *Prop:* $f^{**} = f$ (if proper/closed). $y \in \partial f(x) \iff x \in \partial f^*(y)$.

The 4-Step Recipe: 1. Maximize: $\nabla_x (y^\top x - f(x)) = 0 \implies \nabla f(x) = y$. 2. Solve for x in terms of $y \ (x_{(y)})$. 3. If $x_y \in \text{int}(\text{dom } f)$, $f^*(y) = y^\top x_y - f(x_y)$. 4. **Restricted Domain:** If $x \in [a, b]$, check boundaries. - If $x_y < a$, max is at $x = a$. If $x_y > b$, max is at $x = b$.

Standard Pairs & Quiz Examples:

- $f(x) = -\ln x \rightarrow f^*(y) = -1 - \ln(-y)$ for $y < 0$.
- $f(x) = e^x \rightarrow f^*(y) = y \ln y - y$ for $y > 0$.
- $f(x) = \frac{1}{2} x^\top Q x \rightarrow f^*(y) = \frac{1}{2} y^\top Q^{-1} y$.
- $f(x) = -x^2$ on $[-1, 1] \rightarrow f^*(y) = 1 + |y|$ (V-shape).
- $f(x) = x^2 + x$ on $[-1, 1]$: Stationary $x = \frac{1}{2}(y - 1)$. $f^*(y) = -y \ (y \leq -1); \frac{1}{4}(y - 1)^2 \ (-1 \leq y \leq 3); y - 2 \ (y \geq 3)$.

3. Subdifferentials & Projections

$\partial f(\bar{x}) = \{v \mid f(x) \geq f(\bar{x}) + v^\top (x - \bar{x})\}$.

Optimality: $0 \in \partial f(x^*) \iff x^*$ global min.

- $\partial |x| = \{-1\}$ if $x < 0, \{1\}$ if $x > 0$, and $[-1, 1]$ if $x = 0$.
- Graphical Tactic:** Minimizer is where the subdifferential plot contains the 0-line (look for vertical segments spanning $y = 0$).

Projections (PGD): $x_{k+1} = \text{prox}_{T, C}(x_k - T \nabla f(x_k))$.

- ℓ_∞ -**ball** ($\|x\|_\infty \leq 1$): $x_i = \text{clamp}(z_i, -1, 1)$.
- ℓ_2 -**ball** ($\|x\|_2 \leq 1$): $z / \max(1, \|z\|_2)$.
- ℓ_1 -**ball** ($\|x\|_1 \leq 1$): Soft-thresholding.
- Orthogonal U :** If $C = \{x \mid \|Ux\|_\infty \leq 1\}$, use $x = U^\top \text{prox}_{\|\cdot\|_\infty}(Uz)$.

4. Duality & KKT

Primal: $\min f(x)$ s.t. $g_i(x) \leq 0, h_j(x) = 0$.

Lagrangian: $\mathcal{L} = f(x) + \sum \lambda_i g_i(x) + \nu^\top (Ax - b)$.

Dual Function: $d(\lambda, \nu) = \inf_x \mathcal{L}$. Always concave.

Weak Duality: $d^* \leq p^*$ *always*.

KKT Conditions (Strong duality requires Convexity + Slater's condition: $\exists \bar{x}, g_i(\bar{x}) < 0$): 1. **Stationarity:** $\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0$. 2. **Primal Feas:** $g(x^*) \leq 0, Ax^* = b$. 3. **Dual Feas:** $\lambda^* \geq 0$. 4. **Comp. Slackness:** $\lambda_i^* g_i(x^*) = 0$.

Tactic: Write full $\mathcal{L}$, set $\nabla_x \mathcal{L} = 0$, substitute x into $d(\lambda)$, maximize (often quadratic). Assume constraints active/inactive based on objective push.

5. Optimization Classes

Hierarchy: $LP \subset QP \ (P \succeq 0) \subset QCQP \subset SOCP \subset SDP$.

- SOCP:** $\|Ax + b\|_2 \leq c^\top x + d$.
- Trap:** $|A_i x + b_i|^2 \leq c^\top x + d$ is **not** always QP (if $c \neq 0$, quad constr is not a quadratic form). It is SOCP.

6. Smoothness, Strong Conv. & Rates

Smooth (L): $\nabla^2 f \preceq L I$. **Strong Conv (μ):** $\nabla^2 f \succeq \mu I$. **Condition Number:** $\kappa = L/\mu$.

Setup	Method	Rate / Iters N
L, μ	GD / PGD	$O(\kappa \ln(1/\epsilon))$
L, μ	Nesterov	$O(\sqrt{\kappa} \ln(1/\epsilon))$
L only	GD / Nest	$O(1/\epsilon) / O(1/\sqrt{\epsilon})$
Lipschitz (L_f)	Subgrad	$O(1/\epsilon^2)$

Iteration Bound Recipe: For $\|x_N - x^*\| \leq \epsilon$ with optimal step $T = \frac{2}{L + \mu}$:

$$N \geq \frac{\ln(|x_0 - x^*|/\epsilon)}{\ln((\kappa + 1)/(\kappa - 1))} \approx \frac{\kappa + 1}{2} \ln \frac{|x_0 - x^*|}{\epsilon}$$

7. Hand Calculus & Matrix Hacks

Derivatives: $\nabla(x^\top A x) = (A + A^\top)x, \nabla^2 = A + A^\top$.

Quadratic Form: $f = \frac{1}{2} x^\top Q x + b^\top x + c \implies \nabla^2 f = Q$. Min at $-Q^{-1}b$ (if $Q \succ 0$). $\mu = \lambda_{\min}(Q), L = \lambda_{\max}(Q)$.

Steepest Descent Step: $T_k = \frac{\nabla f_k^\top \nabla f_k}{\nabla f_k^\top Q \nabla f_k}$.

Completing the Square (Vector): $x^\top Q x - 2b^\top x = (x - Q^{-1}b)^\top Q (x - Q^{-1}b) - b^\top Q^{-1}b$.

2 x 2 Eigenvalues (Fast L & μ):

$$\lambda^2 - \text{Tr}(A)\lambda + \det(A) = 0 \implies \lambda_{1,2} = \frac{\text{Tr} \pm \sqrt{\text{Tr}^2 - 4 \det}}{2}$$

$$2 \times 2 \text{ Inverse: } A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Sherman-Morrison (Rank-1 Update):

$$(A + u v^\top)^{-1} = A^{-1} - \frac{A^{-1} u v^\top A^{-1}}{1 + v^\top A^{-1} u}.$$

Schur Complement (Solving KKT Linear Systems):

$$\begin{bmatrix} Q & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \nu \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix}$$

$\Rightarrow$ Solve for $\nu : (AQ^{-1}A^\top)\nu = -(AQ^{-1}c + b)$.
 $\Rightarrow$ Solve for $x : x = -Q^{-1}(A^\top\nu + c)$.